Mini Project: Smart Campus Security System

Multimodal Machine Learning (21AIE541T)

Combining Face Recognition + Scene Understanding + Activity Detection using Multimodal Fusion

1. Project Overview

Scenario: Your university wants a smart security system for campus labs. When a person enters a restricted lab, the system should: (1) Identify WHO the person is, (2) Understand WHAT they are doing, and (3) Determine if this is an authorized activity. You will build this using the multimodal models and fusion techniques learned in Unit 3 and Unit 4.

```
Smart Campus Security Pipeline:

  Camera captures image/frame
         ↓
  ┌──────────────────────────────────────────────┐
  │ Module 1: WHO is this?                        │
  │ InsightFace (ArcFace) → face embedding        │
  │ Compare with enrolled database → identity     │
  ├──────────────────────────────────────────────┤
  │ Module 2: WHAT is happening?                  │
  │ BLIP/Qwen VL → scene caption                  │
  │ 'a person typing on a computer in a lab'      │
  ├──────────────────────────────────────────────┤
  │ Module 3: Is this AUTHORIZED?                 │
  │ CLIP → match scene with allowed activities    │
  │ 'working on computer' ✅  'taking photos' ❌   │
  ├──────────────────────────────────────────────┤
  │ Module 4: FUSION                              │
  │ Combine identity + activity → final decision  │
  │ 'Dhoni is in Lab 3, working on computer' ✅    │
  │ 'Unknown person in Lab 3, taking photos' 🚨    │
  └──────────────────────────────────────────────┘
```

2. Learning Objectives

By completing this project, students will demonstrate:

Concept (Unit)	How It's Applied in This Project
CLIP retrieval (Unit 3)	Match scene descriptions with allowed/disallowed activity labels
BLIP captioning (Unit 3)	Generate text description of the scene from camera image
Encoder-decoder (Unit 3)	BLIP/Qwen VL encoder-decoder generates scene captions
Attention mechanism (Unit 3)	Cross-attention in VLM focuses on relevant image regions
Late Fusion (Unit 4)	Combine face identity score + activity authorization score
Early/Hybrid Fusion (Unit 4)	BLIP ITM or Qwen VL internally fuses image+text features
Face Recognition (Unit 4)	InsightFace identifies people from face embeddings

Biometric system (Unit 4)	Face-based identity verification against enrolled database
Model-free fusion (Unit 4)	Weighted average of identity + activity scores for final decision

3. System Architecture

The system has four modules, each using a different model from the course. The final decision combines all modules through late fusion.

Module	Model Used	Input	Output
1. Face Identity	InsightFace (ArcFace)	Camera image	Person name + confidence score
2. Scene Caption	BLIP / Qwen VL	Camera image	Text description of the scene
3. Activity Check	CLIP	Scene caption + activity labels	Authorized / Unauthorized + score
4. Fusion & Alert	Weighted average	All scores above	Final decision + alert if needed

4. Implementation Steps

4.1 Module 1: Face Identity (InsightFace)

Task: Detect the face in the camera image, extract a 512-dim embedding, and compare with the enrolled face database using cosine similarity.

```
# Enroll faces (run once)
enrolled = {}
for person in ['student_A', 'student_B', 'professor_X']:
    img = cv2.imread(f'enrolled/{person}.jpg')
    faces = app.get(img)
    enrolled[person] = faces[0].embedding

# Identify from camera
camera_img = cv2.imread('camera_capture.jpg')
faces = app.get(camera_img)
query_emb = faces[0].embedding

# Compare with all enrolled faces
for person, emb in enrolled.items():
    score = cosine_sim(query_emb, emb)
    print(f'{person}: {score:.4f}')

# Best match
identity = max(enrolled, key=lambda p: cosine_sim(query_emb, enrolled[p]))
identity_score = cosine_sim(query_emb, enrolled[identity])
```

Deliverable: Enroll at least 5 people (team members). Show the similarity matrix. Demonstrate correct identification with new photos not in the database.

4.2 Module 2: Scene Understanding (BLIP / Qwen VL)

Task: Generate a text description of what is happening in the camera image. Use BLIP for captioning and/or Qwen VL for detailed scene understanding.

```
# BLIP captioning
inputs = caption_processor(images=camera_img, return_tensors='pt').to(device)
output = caption_model.generate(**inputs, max_new_tokens=50)
scene_caption = caption_processor.decode(output[0], skip_special_tokens=True)
# Output: 'a person sitting at a desk with a computer in a laboratory'

# Or use BLIP VQA for specific questions
answer = ask_blip(camera_img, 'What is the person doing?')
# Output: 'typing on a computer'

# Or use Qwen VL for detailed understanding (if available)
# Output: 'A male student wearing a blue shirt is working on a laptop
#          in a computer lab. There are monitors and cables on the desk.'
```

Deliverable: Show captions for at least 10 different campus scenarios (lab work, unauthorized photography, eating in lab, sleeping, using phone, group discussion, etc.).

4.3 Module 3: Activity Authorization (CLIP)

Task: Check whether the detected activity is allowed in the restricted area. Use CLIP to match the scene caption (or image directly) against predefined allowed and disallowed activity labels.

```
# Define allowed and disallowed activities
allowed_activities = [
    'a person working on a computer',
    'a person writing on a whiteboard',
    'a person reading a book',
    'a group of students discussing',
    'a person conducting an experiment',
]

disallowed_activities = [
    'a person taking photos of equipment',
    'a person eating food in the lab',
    'a person sleeping at the desk',
    'an unauthorized person in the lab',
    'a person tampering with equipment',
]

# Use CLIP to score the image against all activities
all_labels = allowed_activities + disallowed_activities
inputs = clip_processor(text=all_labels, images=camera_img, ...)
scores = model(**inputs).logits_per_image.softmax(dim=1)

# Check: is the best match from allowed or disallowed?
best_idx = scores.argmax()
if best_idx < len(allowed_activities):
    activity_status = 'AUTHORIZED'
else:
    activity_status = 'UNAUTHORIZED'
```

Deliverable: Show classification results for all 10 campus scenarios. Show bar charts of CLIP scores for allowed vs disallowed labels. Discuss any misclassifications.

4.4 Module 4: Fusion & Alert System (Late Fusion)

Task: Combine the outputs of all three modules into a final security decision using late fusion.

```
# Decision logic using late fusion

def security_decision(identity, identity_score, scene_caption, activity_status,
activity_score):

    # Case 1: Known person + Authorized activity → OK
    if identity_score > 0.4 and activity_status == 'AUTHORIZED':
        alert = 'GREEN'
        message = f'{identity} is in the lab. Activity: {scene_caption}. All clear.'

    # Case 2: Known person + Unauthorized activity → WARNING
    elif identity_score > 0.4 and activity_status == 'UNAUTHORIZED':
        alert = 'YELLOW'
        message = f'WARNING: {identity} performing unauthorized activity: {scene_caption}'

    # Case 3: Unknown person + Any activity → ALERT
    elif identity_score < 0.3:
        alert = 'RED'
        message = f'ALERT: Unknown person detected! Activity: {scene_caption}'

    # Case 4: Uncertain identity → CHECK
    else:
        alert = 'YELLOW'
        message = f'Uncertain identity (score={identity_score:.2f}). Activity:
{scene_caption}'

    return alert, message
```

Deliverable: Show the full pipeline running end-to-end on at least 10 test images. Display results in a summary table with columns: Image, Identity, Confidence, Scene Caption, Activity Status, Alert Level.

5. Data Requirements

Data	Quantity	How to Collect
Enrolled face photos	5 people × 2-3 photos each	Team members take photos with phone
Campus scenario images	At least 10 diverse scenarios	Take photos in your lab/classroom
Allowed activity labels	5-8 labels	Define based on your lab rules
Disallowed activity labels	5-8 labels	Define based on your lab rules
Test images (query)	10+ images	New photos not in enrolled set

Suggested campus scenarios to capture: Person working on computer, writing on whiteboard, reading, group discussion, eating food in lab, sleeping at desk, using phone, taking photos of equipment, entering empty lab, two people talking, person with backpack, person carrying equipment.

6. Evaluation Criteria

Component	Marks	What We Look For
Module 1: Face Recognition	15	Correct enrollment, similarity matrix, identification accuracy
Module 2: Scene Captioning	15	Meaningful captions for diverse scenarios, BLIP or Qwen VL used
Module 3: Activity Classification	15	CLIP-based scoring, well-designed activity labels, bar charts
Module 4: Fusion & Alert	20	Correct fusion logic, meaningful alert levels, end-to-end demo
Analysis & Discussion	15	Where does the system fail? Why? How could it be improved?
Code Quality & Presentation	10	Clean code, comments, visualizations, organized notebook
Creativity / Extensions	10	Bonus features (see Section 7)
Total	100	

7. Bonus Extensions (Optional)

Students who want to go further can implement any of the following for extra credit:

Extension	Description	Models Used
Hindi Alert Messages	Translate alert messages to Hindi using MarianMT	MarianMT (from Translation notebook)

Attention Visualization	Show which image regions BLIP focuses on when generating captions	BLIP cross-attention weights
Voice Verification	Add voice biometric as additional modality for identity	Resemblyzer (from Face+Voice notebook)
Multiple Fusion Strategies	Compare late vs early vs hybrid fusion for the alert decision	CLIP + BLIP ITM (from Fusion notebook)
Video Processing	Process a short video clip frame-by-frame instead of single image	OpenCV + any VLM
VLM Fine-tuning	Fine-tune Qwen VL on custom campus security descriptions	Unsloth + Qwen3.5

8. Submission Guidelines

Submit a single Jupyter notebook (.ipynb) with:

Item	Requirement
Notebook	All code cells executed with visible outputs
Images	At least 10 test scenario images embedded or loaded from Drive
Enrolled Database	At least 5 people enrolled with face photos
Results Table	Summary table showing all test results (identity, caption, activity, alert)
Analysis	Written discussion of 3+ failure cases and why they failed
Team Info	Team name, member names, roll numbers at the top of notebook

Team size: 2-3 students per team

9. How This Connects to the Course

```
Unit 3 → Unit 4 → Mini Project: Everything comes together!

  CLIP (Unit 3)           → Activity classification (Module 3)
  BLIP captioning (Unit 3) → Scene understanding (Module 2)
  BLIP VQA (Unit 3)       → Answering 'What is the person doing?'
  Translation (Unit 3)    → Hindi alert messages (Bonus)
  Attention (Unit 3)      → Visualize what the model sees (Bonus)
  Face Recognition (Unit 4) → Identity verification (Module 1)
  Late Fusion (Unit 4)    → Combining identity + activity scores (Module 4)
  Biometric System (Unit 4) → The overall security pipeline
  VLM Fine-tuning (Unit 4) → Custom campus descriptions (Bonus)

  This IS a multimodal system:
    Visual (camera image) + Text (captions, labels) + Biometric (face)
    All fused together for one decision.
```

Good luck! Build something impressive.